

Problem V. Chromatic Number

Time Limit	2000 ms
Mem Limit	262144 kB
Code Length Limit	10240 B

You are given a simple undirected connected **graph** with N nodes and M edges with positive costs. From all the paths between 1 and N we'll call those that have the minimum total cost *optimal*. You should *select* K nodes such that the number of optimal paths that pass through all the K selected nodes is maximum.

Standard input

The first line contains 3 integers N , M and K .

Each of the next M lines contains 3 integers A , B and C , representing an edge between nodes A and B having cost C .

Standard output

Print a line containing two integers:

- The first integer is the maximum number of optimal paths that can pass through the K selected nodes
- The second integer represents the number of ways of selecting the K nodes such that the number of optimal paths passing through them is maximum. This value should be printed modulo $10^9 + 7$.

Constraints and notes

- $1 \leq N \leq 300$
- $N - 1 \leq M \leq \frac{N(N-1)}{2}$
- $1 \leq A, B \leq N$
- $1 \leq C \leq 10^9$
- $1 \leq K \leq N$
- It is guaranteed that the first value you need to output is strictly positive and that it fits in a 64 bit signed integer.

Example 1

Input	Output
4 5 3 1 3 2 1 2 1 2 3 1 2 4 2 3 4 1	2 2

The distance between 1 and 4 is 3.

There are 3 possible paths of minimum cost from the node 1 to node 4:

$\{(1, 2, 4), (1, 3, 4), (1, 2, 3, 4)\}$.

You can choose the 3 special nodes as $\{1, 2, 4\}$, then the 2 following optimal paths will pass through all of them: $\{(1, 2, 4), (1, 2, 3, 4)\}$.

You can also choose the 3 special nodes as $\{1, 3, 4\}$. In this case, the following 2 paths will pass through all of them: $\{(1, 2, 3, 4), (1, 3, 4)\}$.

Example 2

Input	Output
8 10 2 1 2 3 1 3 5 2 4 4 4 3 2 4 5 1 5 6 3 6 8 1 5 7 2 7 6 1 7 8 2	6 6

The distance from 1 to 8 is 12.

There are 6 paths of such cost:

1. (1, 2, 4, 5, 7, 8)
2. (1, 3, 4, 5, 7, 8)
3. (1, 2, 4, 5, 7, 6, 8)
4. (1, 3, 4, 5, 7, 6, 8)
5. (1, 2, 4, 5, 6, 8)
6. (1, 3, 4, 5, 6, 8)

You can choose the following subsets of 2 nodes:

$\{\{1, 4\}, \{1, 5\}, \{1, 8\}, \{4, 5\}, \{4, 8\}, \{5, 8\}\}$.